

Cinema Booking Chatbot

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Booking tickets for entertainment is being revolutionized by the Cinema Booking Chatbot, a cutting-edge use of Natural Language Processing (NLP) and Artificial Intelligence (AI). Lemmatization and tokenization, two sophisticated NLP techniques, are used by the chatbot to provide an interactive platform for easy movie ticket purchases. Users can easily express their preferences, explore movie options, choose showtimes, and manage reservations thanks to a strong intent-matching mechanism. A sophisticated understanding of natural language is ensured by the system through its sophisticated tokenization and lemmatization processes. Furthermore, a clever recommendation engine improves the user experience by making personalized recommendations for well-liked movies. The chatbot expedites the booking process by gathering essential information quickly and offering an intuitive user interface. In addition to ticketing, it answers questions and provides support, acting as an educational resource. The Cinema Booking Chatbot represents a revolution in the entertainment industry.

Additional Keywords and Phrases: Natural Language Processing, Intent matching, tokenization and lemmatization, Question Answering, performing tasks.

1 INTRODUCTION

Introducing the cutting-edge Cinema Booking Chatbot, a clever system created to completely transform the process of booking movie tickets. Using state-of-the-art Natural Language Processing (NLP) methods, the chatbot offers consumers a simple, interactive way to purchase movie tickets. By using sophisticated tokenization and lemmatization, the system can interpret user inputs and ensures that it is aware of the subtleties inherent in natural language. Users can easily communicate their preferences, find out what movies are available, choose showtimes, and even modify or cancel reservations thanks to a strong intent-matching mechanism.

To provide a unique and interesting experience, the chatbot integrates a smart recommendation engine that makes recommendations for well-liked films based on user preferences. By asking users for important details like movie preference, showtime, and number of tickets, it streamlines the booking process. The system also responds to a variety of questions, giving users access to movie times, theatre locations, and support for any issues they might be having. The Cinema Booking Chatbot seeks to improve user experience, expedite ticket bookings, and usher in a new era of convenience in the entertainment booking industry with its intuitive interface and conversational capabilities. Welcome to our NLP-powered chatbot's smarter approach to buying movie tickets!

2 CHATBOT ARCHITECTURE

With the use of natural language processing (NLP), the versatile and user-friendly Cinema Booking Chatbot was created to streamline the process of booking movie tickets. The chatbot's core functionality is its capacity to comprehend and react

to user input concerning movie reservations, thereby offering a smooth and interactive experience. Through a conversational interface, users can choose showtimes, make movie inquiries, and take actions like making, changing, or canceling reservations.

2.1 Chatbot Architecture Flowchart

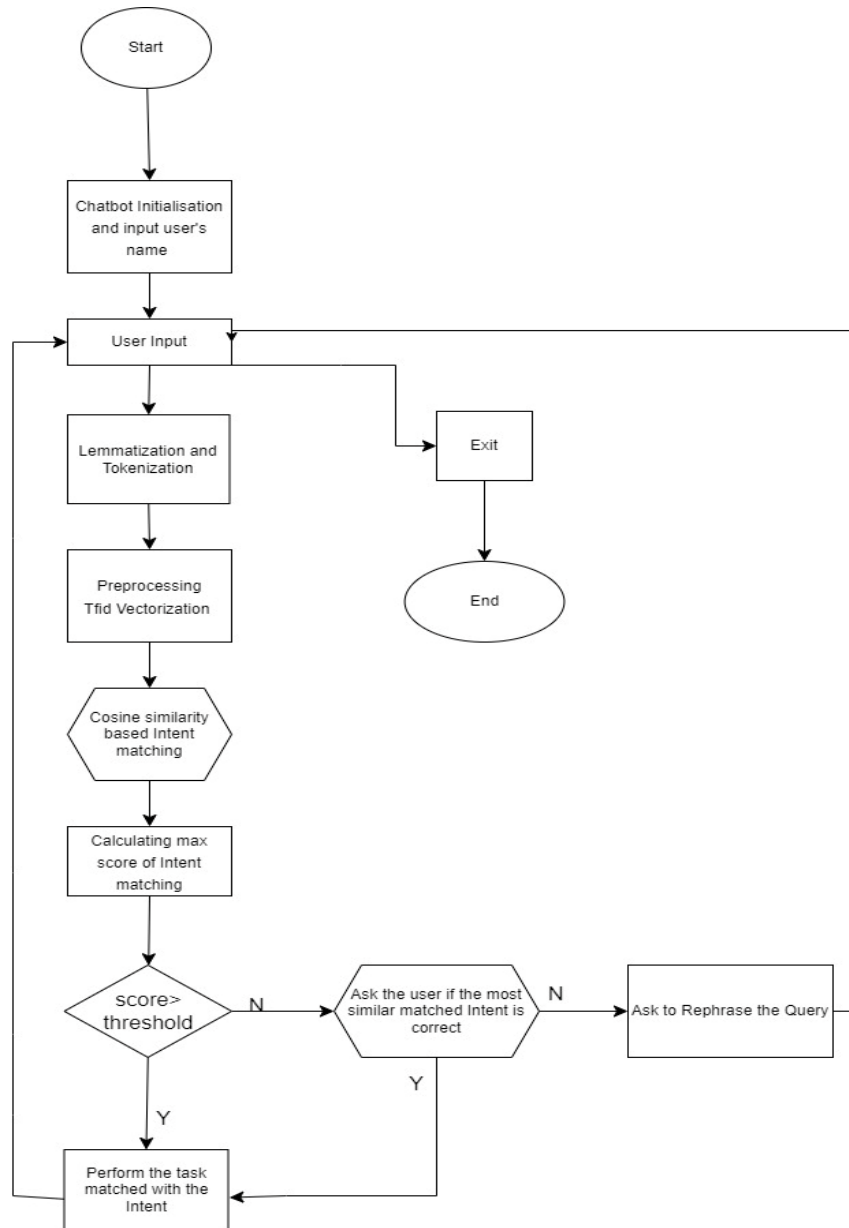


Figure 1: Flowchart of Architecture of chatbot

The below steps describe the flow of the chatbot system and how it works.

Step 1: Start

Step 2: Chatbot Initialisation

The chatbot initiates the conversation and asks for the user's name. The inputs their name and the chatbot greets the user.

Step 3: User Input

The system asks for the user query.

Step 4: Lemmatization and Tokenization

The system lemmatizes and tokenizes the user's query. This means that the system converts all of the words in the query to their base forms and splits the query into individual tokens.

Step 5: Preprocessing TfId Vectorization

The system preprocesses the data and uses TfId vectorization to further match with the dataset.

Step 6: Intent Matching

The system uses cosine similarity to match the user's query to a set of known intents. An intent is a representation of what the user wants to achieve with their query.

Step 7: Calculating max score

The system calculates the maximum score of the matched intent, if it's greater than the threshold value it goes to step 8.

Step 8: Not Matched

If the system cannot match the user's query to any known intents, it asks the user if the most similar matched intent is correct. If it is correct it goes to the next step else it asks the user to rephrase their query and the process again starts from Step 3.

Step 9: Perform the Task

If the system matches the user's query to a known intent, it performs the task associated with that intent. For example, if the user's query is "I want to book cinema tickets?", the system calls the booking function and books the movie ticket for the user. It goes back to step 3.

Step 10: Exit

2.2 Implementation

2.2.1 Intent Matching

```
#Cosine similarity based Intent matching to give best matching intent response
def get_intent(user_input):
    lemmatized_input = " ".join(tokenize_and_lemmatize(user_input))
    user_vector = vectorizer.transform([lemmatized_input])
    similarities = cosine_similarity(user_vector, intent_vectors)
    most_similar_intent_index = similarities.argmax()
    return intent_labels[most_similar_intent_index], similarities[0, most_similar_intent_index]
```

Figure 2: Intent matching

The get-intent function takes the user input, preprocesses it by lemmatizing and tokenizing, converts it into a vector, calculates the cosine similarity with pre-defined intent vectors, and finally returns the label of the most similar intent along

with the similarity score. This way, the system can determine the user's intent based on the provided intents and responses in your chatbot. Identity Management

```
#Initializing the user input and retrieving the name
username = None
if username is None:
    username = input("Chatbot: Hello!welcome to cinema booking! What's your name?\n(you can type exit to leave:/)")
    print(f"Chatbot: Nice to meet you, {username}!\n")
    print('Chatbot:You can type help to know what all you can do')

#Main while loop for the chatbot
while True:
    user_input = input("User: ")
    #Case for if user asks for their name
    if user_input.lower() == 'what is my name?':
        print(f'Chatbot:your name is {username}')
        continue
```

Figure 3: Identity management

First, it checks if the username is None, and if so, it prompts the user to enter their name. Then, if the system is prompted for a username it prints the username.

2.2.2 Transaction

```
def cancel_booking(book_id):
    ans=input('Have you booked tickets?(Y/N)')
    if ans.lower()=='y':
        cancelled_id = input('Chatbot:Please enter the booking id: ')
        if cancelled_id == book_id:
            print(f"Chatbot:Booking {book_id} has been canceled.")
        else:
            print(f"Chatbot:Booking ID not found.")
    else:
        print('Please book the tickets first')
```

Figure 4: Transaction

The function prompts the user to confirm if they have booked tickets and then asks for the booking ID to be canceled. This function ensures that the user has booked tickets before proceeding with the cancellation process. If the user has booked tickets and provides the correct booking ID, it cancels the booking; otherwise, it indicates that the booking ID is not found.

2.2.3 Information Retrieval and Question-Answering

```
def calculate_similarity(user_input, qa_data):
    vectorizer = TfidfVectorizer()
    processed_user_input = preprocess(user_input)
    corpus = qa_data['Processed_Question'].tolist() + [processed_user_input]
    tfidf_matrix = vectorizer.fit_transform(corpus)
    similarity_scores = cosine_similarity(tfidf_matrix[-1], tfidf_matrix[:-1])
    return similarity_scores[0]
```

Figure 5: Information retrieval and question-answering

The function calculates the similarity between a user input and a set of processed questions and answers (QA data). This similarity is computed using the TF-IDF (Term Frequency-Inverse Document Frequency) representation and cosine similarity. The resulting similarity score is an array where each element represents the similarity between the user input and the corresponding question.

2.2.4 Small Talk

```
#User Intents and the responses given by chatbot for small talk
intents = {
    "greeting": ["hello", "hi", "hey", "howdy"],
    "goodbye": ["bye", "goodbye", "see you"],
    "thanks": ["thank you", "thanks a lot", "thanks"],
    "job": ["What is your job", "what is your work", "what can you do"],
    "feeling": ["How are you today", "How are you"],
    "good": ["I am good too", "I feel fine", "Good!", "Fine", "I am good", "I am great", "great", "good"],
    "bad": ["Bad", "Not good", "Not well", "Not fine", "Not feeling well"],
}

responses = {
    "greeting": ["Hello!", "Hi there!", "Hey! How can I help you?"],
    "goodbye": ["Goodbye!", "See you later!", "Have a great day!"],
    "thanks": ["You're welcome!", "No problem!", "Anytime!"],
    "job": ["I can help you book cinema tickets", "I can show you list of cinemas and you can book"],
    "feeling": ["I am feeling good, you?", "Very good and you?", "Actually, I'm okay and you?"],
    "good": ["That is perfect!"],
    "bad": ["Oh no! feel better", "Hope you feel better", "Sorry to hear that"]
}
```

Figure 6: Small talk

For small talk a pre-defined intents and corresponding responses for a chatbot is created. This is a common structure used in natural language processing (NLP) to create a basic question-and-answer system. Users' inputs are categorized into intents, and the chatbot responds based on the identified intent.

2.3 Functionalities of chatbot

2.3.1 Movie Ticket Booking

```
User: I want to book
Chatbot:Wohoo!Lets get started!
Chatbot:Here is a list of all the cinemas
    KGF
    RRR
    Home Alone
    Avengers
    Iron Man
    Spider Man
    Grinch

Chatbot:Please give me a cinema name: KGF
Chatbot:That is a good choice
Chatbot:Here is the list of theatres
    Cinemaworld
    Inox
    PVR
    Victoria Theatre

Chatbot:Enter the theatre name: Inox
Chatbot:Choose a time from the below
    12PM
    3PM
    9PM:
    12PM
Chatbot:you are almost there!

Chatbot:Which date do you want to see the cinema?
    2
Chatbot:Alright 2 it is!

Chatbot:Please enter the number of tickets: 4
Chatbot: cinema is: KGF,and 4 Tickets at12PM
Chatbot:Booking successful! Your Booking ID is 92372.
Chatbot:Enjoy the cinema Varsha:)
```

Figure 7: Movie ticket booking

Figure 7 is an example of how the booking works. First, the user inputs the query to book tickets, and the chatbot responds with a list of movies and asks the user to input the movie name. The same process is followed for choosing the theatre,

time, date, and the number of tickets. When all the steps are completed the system confirms by providing a booking ID and ends by greeting the user.

2.3.2 Cancel Tickets

```
User: I want to cancel tickets
Have you booked tickets?(Y/N)Y
Chatbot:Please enter the booking id: 92372
Chatbot:Booking 92372 has been canceled.
```

Figure 8: Cancel tickets

Figure 8 is an example of how canceling ticket works. First, the user inputs the query to cancel the tickets, the chatbot confirms first if the tickets have been booked or not. If yes then ask the user for booking id. If the booking id matches then it proceeds to cancel the booking.

2.3.3 Update The Tickets

```
User: I want to change the tickets
Chatbot:Enter the booking ID92372
Chatbot:Enter the new cinema name: RRR
Enter the theatre nameInox
Chatbot:Enter the new number of tickets: 2
Chatbot:Enter the new time (12PM, 3PM, 9PM): 12PM
Chatbot:Here is your updated information! cinema is: RRR, Tickets: 2, Time: 12PM
```

Figure 9: Update the tickets

Figure 9 is an example of how updating the tickets works. First, the user inputs the query to make changes in the ticket, the chatbot asks for the booking id and then proceeds to ask the user for a new movie name, theatre, time, and number of tickets.

2.3.4 Listing The Cinemas

```
User: list the cinemas
Chatbot:KGF
RRR
Home Alone
Avengers
Iron Man
Spider Man
Grinch
```

Figure 10: Listing the cinemas

Figure 10 is an example of listing the movies. First, the user inputs the query to list the movie names. The chatbot lists out all the current playing movies.

2.3.5 Listing The Theatres

```
User: what theaters are showing cinema right now
Chatbot: Here is the list of theatres
Cinemaworld
Inox
PVR
Victoria Theatre
```

Figure 11: Listing the theatres

Figure 11 is an example of listing the theatres. First, the user inputs the query to list the theatres showing the movies. The chatbot lists out all the theatres that are showing the movies.

2.3.6 Picking A Cinema For User

```
User: can you pick a cinema
How about Home Alone cinema?
User: thanks
Chatbot: No problem!
```

Figure 12: Picking a cinema for the user

Figure 12 is an example of picking a movie. First, the user inputs the query to pick a movie. The chatbot responds by choosing a movie from the available list.

2.4 Justification

Several essential elements were involved in the chatbot's implementation. To match user queries with predefined intents and responses, the system first used the scikit-learn library for machine learning tasks. This included implementing a TF-IDF vectorizer and calculating cosine similarity. To improve the chatbot's comprehension of natural language, the NLTK library was utilized in the lemmatization and tokenization process.

Because TF-IDF and cosine similarity were so good at capturing the semantic similarity between user inputs and predefined intents, that was the rationale behind using them. With this method, the chatbot can respond to user inquiries in a contextually relevant manner. To increase user engagement, the code also included a dynamic movie recommendation engine that makes recommendations for well-known movies.

The efficacy of NLP techniques in managing conversational interactions serves as justification for this implementation. The computation of cosine similarity and TF-IDF vectorization improve the system's ability to accurately interpret user intent. By standardizing words, the lemmatization process helps the chatbot understand language better. All things considered, the selected methodology complies with NLP best practices, guaranteeing that users receive a solid and user-friendly movie booking experience from the chatbot.

3 CONVERSATIONAL DESIGN

A number of crucial techniques were used in the Cinema Booking Chatbot's implementation of Conversational Design Principles to improve user experience and promote efficient communication. The creation of organic, user-centered, and

context-aware interactions is the main goal of these guidelines. An outline of how these ideas were applied is provided below:

3.1 Prompt Design

```
User: list all the cinema
Chatbot:KGF
RRR
Home Alone
Avengers
Iron Man
Spider Man
Grinch
```

Figure 13: Prompt Design

Figure 13 refers to prompt design. When the user inputs “lists”, the chatbot lists all the cinema names and asks the user to input a choice. It supports the user query and performs the task.

3.2 Discoverability

```
Chatbot: Hello!Welcome to cinema booking! What's your name?
(you can type exit to leave:/)varsha
Chatbot: Nice to meet you, varsha!

Chatbot:You can type help to know what all you can do
User: help
Here is some of the things you can type
1.Show me the list of cinema
2.I want to book cinema tickets
3.I want to cancel the cinema tickets
4.I need to change the tickets
5.List of cinemas
6.Show me list of theatres
7.Hi/Hello
```

Figure 14: Discoverability

Figure 14 refers to discoverability. The chatbot lets the user know to type help. When the user types help it shows example prompts the user can give. Whether requested or provided explicitly, the system offers the discoverability option.

```
User: what can you do
Chatbot: I can help you book cinema tickets
```

Figure 15: Discoverability

Figure 15 shows one more example of the discoverability. Users can simply type What can you do?

3.3 Error Handling

When a user poses an unclear question or makes a mistake, the chatbot provides helpful feedback to point them in the direction of the right input.

```
User: i want to book tickets
Chatbot:Wohoo!Lets get started!
Chatbot:Here is a list of all the cinemas
    KGF
    RRR
    Home Alone
    Avengers
    Iron Man
    Spider Man
    Grinch

Chatbot:Please give me a cinema name: sdf
Chatbot:Please select from cinemas available only!
```

Figure 16: Error Handling

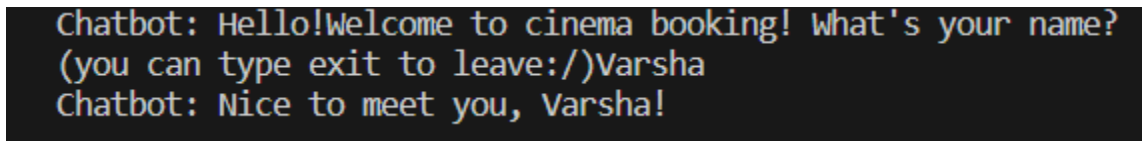
Figure 16 refers to Error Handling. When the user inputs a movie other than the available movies, it requests the user to select the movie from the available list.

```
User: blah blah
Chatbot: I'm sorry, I didn't understand that.
User: fnsnfs
Chatbot: I'm sorry, I didn't understand that.
```

Figure 17: Error Handling

Figure 17 shows another example of error handling. When the user inputs gibberish it responds by saying I didn't understand.

3.4 Personalization

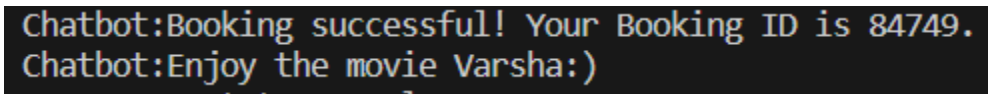


```
Chatbot: Hello!welcome to cinema booking! what's your name?  
(you can type exit to leave:/)Varsha  
Chatbot: Nice to meet you, Varsha!
```

Figure 18: Personalization

Figure 18 refers to the chatbot ensuring that users are greeted with kindness and warmth by beginning conversations with salutations. The chatbot might say, for example, "Hello! Greetings from Cinema Booking The system makes an exchange more engaging and tailored by referring to users by the name they have supplied. It asks for the username first.

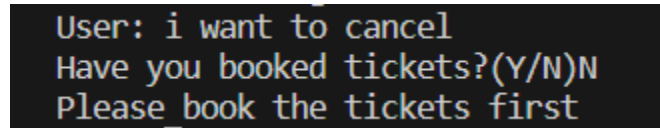
3.5 Confirmation



```
Chatbot:Booking successful! Your Booking ID is 84749.  
Chatbot:Enjoy the movie Varsha:)
```

Figure 19: Confirmation

Figure 19 refers to the confirmation given by the chatbot by providing the booking ID. Confirmation Messages: When a user completes a successful action, such as scheduling or canceling a ticket, the chatbot provides clear confirmation messages, which increase user confidence.

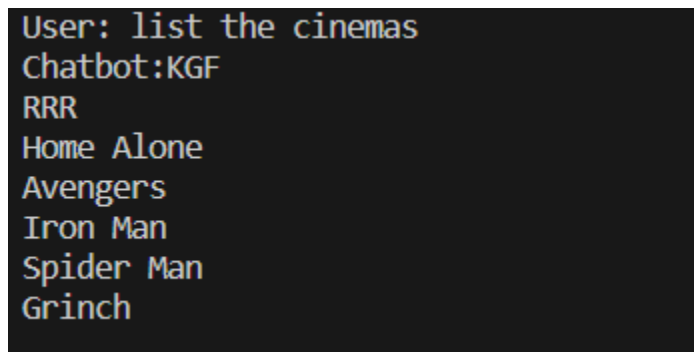


```
User: i want to cancel  
Have you booked tickets?(Y/N)N  
Please book the tickets first
```

Figure 20: Confirmation

Figure 20 refers to Confirmation. When the user wants to cancel the tickets, Chatbot first confirms if the user has booked the tickets or not. If not, then it asks the user to book the tickets first.

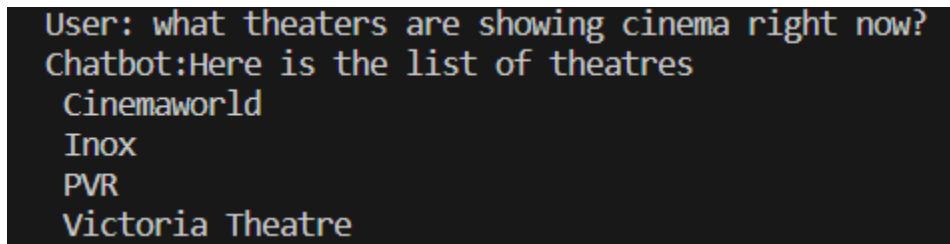
3.6 Context



```
User: list the cinemas  
Chatbot:KGF  
RRR  
Home Alone  
Avengers  
Iron Man  
Spider Man  
Grinch
```

Figure 21: Context

Figure 21 refers to detecting the user intent and appropriately answering. During the conversation, the chatbot remembers user queries and gives relevant responses to queries. Structured Responses

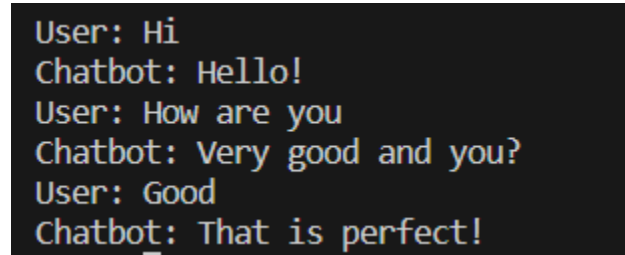


```
User: what theaters are showing cinema right now?  
Chatbot: Here is the list of theatres  
Cinemaworld  
Inox  
PVR  
Victoria Theatre
```

Figure 22: Structured responses

Figure 22 refers to structured responses that are well-organized and succinctly communicate information in an uncomplicated way. The above is an example of clear prompts given by the chatbot to simplify user experience.

3.7 Small Talk



```
User: Hi  
Chatbot: Hello!  
User: How are you  
Chatbot: Very good and you?  
User: Good  
Chatbot: That is perfect!
```

Figure 23: Small talk

Figure 23 refers to the chatbot small interaction with the user. By responding to non-transactional inquiries, the chatbot engages in small talk to foster a more enjoyable and conversational user experience.

With the help of these Conversational Design Principles, the Cinema Booking Chatbot hopes to offer consumers a simple, entertaining, and user-friendly way to make movie reservations. In line with conversational AI best practices, the emphasis on context awareness and natural language understanding makes interactions more smooth and human-like.

4 EVALUATION

The evaluation of the cinema booking chatbot was done with the help of 7 users with different races, genders, and different backgrounds of study.

4.1 Performance Testing

In this testing method, each of the user was given the same set of tasks to complete which included,

- Small talk- Hi/Hello, How are you
- Booking tickets
- Cancelling tickets
- Updating booking
- Movie recommendation

Each of them was given an ID number to maintain the privacy of the user. The aim was to measure errors per scenario encountered the users. The results were as follows.

Percentage of error= (No. of errors/No. of scenarios) *100

ID	No. of errors	No. of scenarios	Error %
1	3	20	15
2	2	15	13.33
3	5	40	12.5
4	6	23	26
5	4	18	22.22
6	4	19	21.05
7	2	16	12.5

Table 1: Error Percentage

4.2 Usability Testing

In this testing method, each of the users was asked to fill out a Google form. The following responses were obtained.

Questions	Agree	Neutral	Disagree
Chatbot was friendly	80%	20%	0%
Chatbot greeted you and asked for your name	100%	0%	0%
Chatbot prompts were clear and concise.	70%	20%	10%
Did not have difficulty navigating	60%	30%	10%
Was able to make small talk	80%	20%	0%
Was able to easily book the tickets	70%	20%	10%
Was able to easily cancel the tickets	60%	30%	10%
Was able to easily update the tickets	70%	30%	0%
Chatbot recommended a nice movie	50%	30%	20%
Chatbot was easy to use	70%	20%	10%

Table 2: User Feedback

Overall, the chatbot was well perceived by the user and some of the key points were,

Positive feedback was particularly noted for the chatbot's ability to handle a wide variety of inputs and its skill in handling errors.

The chatbot's prompts were well-received by users, who appreciated its friendly initial tone, clear explanations of its goals and scope, ease of use, and lack of complexity.

Users were happy with the chatbot's helpfulness, transactional procedures, friendliness, and ability to provide pertinent answers.

5 DISCUSSION

The framework of Responsible Research and Innovation (RRI) is an essential component in the ongoing enhancement and moral evaluation of technological solutions. A thorough examination of the RRI framework about the Cinema Booking Chatbot encompasses several crucial elements.

Engagement with users:

It was crucial to include potential users, students, and other stakeholders like influencers in the chatbot's design and testing stages. Their feedback ensures that the chatbot meets the various needs and preferences of its users by offering insightful information about how to make the system more user-friendly.

Ethical Considerations:

The chatbot complies with privacy laws, guaranteeing the safe handling of user data. Ethical principles are aligned with transparent communication regarding data usage and giving users control over their information.

Inclusivity and Accessibility:

The chatbot was created with inclusivity in mind, taking into account a range of user demographics. To guarantee that a wide range of users can effortlessly engage with the system, accessibility features were incorporated, including a gender-neutral design and an intuitive user interface.

Reflecting on Results:

Through usability testing, the chatbot's effectiveness in facilitating cinema bookings and providing information was evaluated. Performance testing assessed its responsiveness and scalability. Reflecting on the results, any identified issues were addressed, contributing to the iterative development process.

Application Area Impact:

The use of chatbots for movie ticket booking is in line with consumer preferences for effectiveness and convenience. Its effects are seen in the entertainment industry where it expedites the booking process, provides pertinent information, and improves the user experience overall.

Future work:

Including multimodal interaction so that users can communicate with the chatbot via voice, image, or text in addition to text. This might improve the user interface and increase the chatbot's accessibility.

Providing the chatbot permission to communicate with users on social media sites to broaden its reach and offer a consistent experience across various platforms.

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